

Molecular-templated pre-assembly of self-assembled monolayer for perovskite solar cells and modules with improved reverse-bias stability

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Perovskite solar modules, particularly those using ultrathin self-assembled monolayer (SAM)-based hole transport layers, suffer from reverse-bias instability. Here we identified that discontinuous SAM distribution causes shunting and a lower breakdown voltage, while indium tin oxide-triggered electrochemical deprotonation of formamidinium ions leads to reduced long-term stability under reverse-bias conditions. To address these issues, we developed a molecular-templated pre-assembly strategy driven by hydrogen-bonding interactions between the SAM and a polycarbazole template. This approach ensures homogeneous clusters in solution and strong substrate interactions, yielding dense and uniform layers. Subsequently, we prepared minimodules with 24.0% efficiency (certified steady-state efficiency of 23.2%) and improved reverse-bias stability. Small-area devices retained 95% efficiency after 300 h at -4.8 V, while minimodules exhibited a T_{95} lifetime of 312 h under negative open-circuit voltage stress. We showed that a single bypass diode can protect at least 16 subcells, setting a new reliability benchmark for scalable perovskite photovoltaics.

Perovskite photovoltaics have advanced over the past 15 years, evolving from small-area laboratory devices to large-area modules. The field is shifting beyond proof-of-concept demonstrations of high power conversion efficiencies (PCEs) towards scalable module technologies that integrate high efficiency with long-term operational stability. Module-level characterization provides a more representative framework for evaluating manufacturing scalability and enables systematic investigation of stability pathways under realistic operating conditions.

While certified PCEs exceeding 27% have been achieved in small-area perovskite solar cells (PSCs)¹, limited efforts have been

directed towards understanding the stability challenges associated with large-area modules under practical stress conditions, including high-temperature open-circuit bias^{2,3}, high-temperature maximum power point tracking (MPPT) operation^{4–6} and prolonged outdoor exposure⁷. Notably, reverse-bias stability, which is critical for ensuring long-term device reliability, remains underexplored at the module scale.

Reverse-bias instability arises in series-connected modules when partially shaded subcells are compelled to dissipate power from other irradiated subcells⁸. Previous investigations have shown that

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ion migration (for example, iodide ions) and oxidized metal critically undermine the reverse-bias tolerance of PSCs^{9–13}, underscoring the need for compact interfacial layers atop the perovskite film and robust electrodes^{14–16} to increase device breakdown voltage and long-term robustness. Nonetheless, for devices incorporating ultrathin self-assembled monolayers (SAMs) as the hole transport layer (HTL) to achieve record efficiencies^{17–19}, the fundamental degradation processes under reverse bias remain largely unexplored. A recent study showed that SAM-based devices exhibited a markedly reduced breakdown voltage and decreased efficiencies during reverse-bias ageing. Replacing the SAM with a much thicker poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA) layer could boost the robustness under reverse bias¹⁴, highlighting the critical role of compact HTLs for improved reverse-bias stability. However, a practical HTL that achieves an optimal balance between stability and efficiency at the module scale has yet to be realized. Although thick PTAA layers improve stability, they hinder hole extraction and transport, limiting large-area module performance. In addition, the degradation mechanisms and electrochemical reactions at the buried interface under reverse-bias stress remain poorly understood, posing further challenges to the development of robust, high-efficiency perovskite modules.

In this work, we identified discontinuous SAM coverage as the primary cause of the low breakdown voltage and demonstrated that indium tin oxide (ITO)-triggered electrochemical deprotonation of formamidinium (FA⁺) limits the long-term stability of SAM-based PSCs. Through simulations and experiments, we confirmed the electrochemical instability of FA⁺ and elucidated the FA⁺ degradation pathway through continuous N–H cleavage. To address these issues, we applied a molecular-templated pre-assembly strategy driven by hydrogen-bonding interactions, in which the SAM molecules interact with a polycarbazole template to generate homogeneous solution associates and strengthen cluster–substrate interactions. This approach enables a denser and more homogeneous SAM coverage, suppressing shunting pathways and FA⁺ degradation at the buried interface. We designed SAM-based perovskite minimodules that can deliver a maximum PCE of 24.0% with a certified steady-state efficiency of 23.2%. This approach also significantly enhances the reverse-bias stability of devices under accelerated International Summit on Organic Solar Cell Stability protocol for electrical-bias stability testing in the dark at ambient temperature under ambient atmosphere (ISOS-V-1) testing protocols: our small-area cells retained 95% of their original PCE under –4.8 V stress for 300 h and our minimodules maintained 98% efficiency under negative open-circuit voltage ($-V_{oc}$) for 312 h. Furthermore, we also found that a single bypass diode can effectively protect at least 16 subcells under practical operating conditions.

Results and discussion

Reverse-bias degradation mechanism at the buried interface

The ITO/SAM/perovskite/fullerene (C₆₀)/SnO_x/gold (Au) architecture illustrated in Fig. 1a was employed to investigate the degradation mechanism at the buried interface under reverse bias. [4-(3,6-Diphenyl-9H-carbazol-9-yl)butyl]phosphonic acid (Ph-4PACz), previously reported as an HTL for high-efficiency large-area PSCs, was used as the SAM layer¹⁸, FA_{0.9}CS_{0.1}PbI₃ was used as the perovskite absorber to achieve high efficiency^{18,20} and the layers atop the perovskite (C₆₀, SnO_x and Au) have previously been demonstrated to be robust against reverse bias^{14,15} while not altering device efficiency. Considering the morphology of the SAM layer and its possible reactions with the perovskite and bottom electrode, we hypothesized an electrochemical degradation pathway at the buried interface of PSCs under reverse bias (Fig. 1a): due to the discontinuous coverage of the SAM layer²¹, reaction-active regions exist where the perovskite directly contacts the underlying ITO electrode. Electrons can be injected freely into the perovskite, thereby generating an increased reverse current. Electrochemical reactions initiated from the top electrode were ruled out as Au is highly inert compared with Ag

or Cu within our voltage test range (above –5 V)¹⁴. Moreover, cations in the perovskite, that is, Pb²⁺, Cs⁺ and FA⁺, are driven towards the ITO by the reverse-bias electric field. We calculated the migration barriers of these cations, finding that FA⁺ exhibits the lowest barrier and thus migrates most readily under the applied field (Supplementary Fig. 1 and Supplementary Note 1). Among these cations, FA⁺ was also identified as the most probable candidate to undergo electrochemical reactions at the buried interface as it is prone to deprotonate upon contact with oxide surfaces^{22,23}. The resulting protons can then be readily reduced to neutral hydrogen (H⁰) by the injected electron²⁴, further facilitating the deprotonation and accelerating the degradation of the perovskite at the buried interface.

To demonstrate the degradation pathway that we proposed, the breakdown voltage (V_b) and dark current density of PSCs with different HTLs on ITO were first measured to clarify the correlation between HTL coverage and reverse-bias stability (Fig. 1b,c). Four architectures, with a progressively increasing HTL coverage on ITO, were compared: no HTL (perovskite in direct contact with ITO), an ultrathin SAM layer, a thin PTAA layer (2 mg ml^{–1}) and a thick PTAA layer (10 mg ml^{–1}). The corresponding device performances are shown in Supplementary Fig. 2. V_b measurements from +1.2 V to –20.0 V showed that all devices exhibited V_b values below –10 V (Fig. 1b), which is attributed to the robust upper charge transport layer (SnO_x)¹⁵ and chemically stable electrode (Au)¹⁴. Even so, the HTL-free devices demonstrated the lowest V_b (–10.8 V), whereas incorporation of the SAM, 2 mg ml^{–1} PTAA and 10 mg ml^{–1} PTAA led to a progressive decrease in V_b from –12.0 V to –13.5 V and –18.0 V, respectively, consistent with a previous report¹⁴. Dark current measurements (Fig. 1c) revealed that all four device architectures exhibited similarly low values (<10^{–3} mA cm^{–2}) at voltages above –2 V. Upon decreasing the reverse bias to –5 V, the HTL-free, SAM-based and thin PTAA-based devices showed a pronounced increase in dark current by one-to-two orders of magnitude, with the HTL-free device exhibiting the highest value (0.05 mA cm^{–2}). In contrast, devices incorporating the most compact HTL (10 mg ml^{–1} PTAA) retained low dark currents (6.5 × 10^{–4} mA cm^{–2}), even under high reverse-bias conditions. These results indicate that improved HTL coverage positively correlates with reduced dark currents at elevated reverse voltages. This is because the HTL plays a critical role in blocking electron injections from ITO to the perovskite under reverse bias, while a discontinuous HTL (such as a SAM and thin PTAA layer) would lead to potential current leakage pathways at the buried interface, especially at high reverse bias.

Next, we investigated the potential reactions at the buried interface by ageing HTL-free and SAM-based devices at –4.8 V for 300 h. The perovskite films were then peeled from the ITO substrates using a previously reported method³ and the exposed bottom surfaces were characterized by X-ray photoelectron spectroscopy (XPS) and compared with fresh HTL-free samples. No significant changes were observed in the Pb and Cs signals (Supplementary Fig. 3); however, an additional peak corresponding to deprotonated FA was detected in the N 1s spectra of both the aged HTL-free and SAM-based devices (Fig. 1d and Supplementary Fig. 4a)^{22,25,26}. The deprotonated FA fraction was higher in the HTL-free device (13.6%) than in the SAM-based device (9.0%), indicating that the SAM layer partially blocks electrons, inhibiting the deprotonation of FA⁺. However, it cannot fully prevent electron injection from ITO, leading to the partial degradation of FA⁺ (Supplementary Fig. 4b). Furthermore, ¹H nuclear magnetic resonance (¹H NMR) spectroscopy (Supplementary Fig. 5) revealed that the reverse-bias-aged HTL-free samples showed a reduction in the integral area of the N–H peak, associated with an alteration in signal shape. In contrast, no significant change was observed in the C–H peak. This further confirms the deprotonation of FA⁺ upon electron injection and subsequent proton reduction under reverse bias. We also conducted operando photoluminescence on the SAM-based device under reverse bias (Supplementary Fig. 6). Initially, the photoluminescence peak was intense and centred at 800 nm. During ageing, the emission band

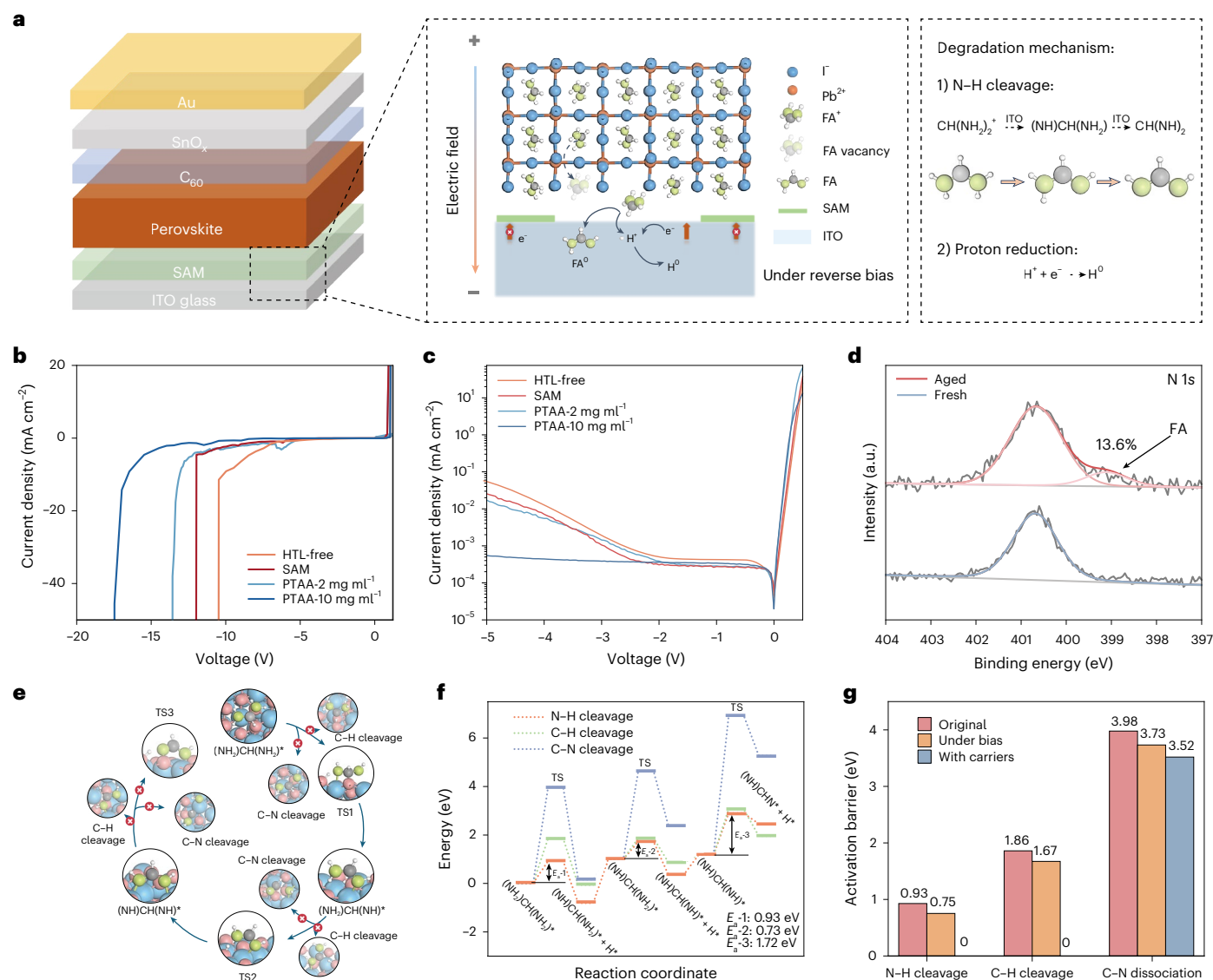


Fig. 1 | Degradation mechanism at the buried interface under reverse bias.

a, Diagram illustrating the degradation mechanisms at the buried interface of PSC. The inset shows ITO/SAM/Perovskite buried interface under reverse bias. **b**, J - V curves (from 1.2 V to -20.0 V) of PSCs with different HTLs. **c**, Dark J - V curves of PSCs with various HTLs. **d**, XPS N 1s spectra of perovskite films peeled off a fresh and aged (-4.8 V for 300 h) device without an HTL (curves fitted by XPSPEAK, linear background; the arrow acts as a guide to the eye). **e**, Scheme

showing the dominant degradation pathway of FA^+ , as well as the transition states for each elementary reaction step. Blue spheres denote In atoms, red spheres denote O atoms, green spheres denote N atoms, grey spheres denote C atoms and white spheres denote H atoms. TS, transition state. *, adsorption state of molecules. **f**, Potential energy diagram for the degradation reactions of FA^+ . **g**, Activation barriers for the possible degradation reactions of FA^+ under bias and with accumulated carriers.

progressively broadened and experienced a blueshift, indicating a gradual loss of FA^+ and perovskite decomposition (Supplementary Note 2). The slight increase of the PbI_2 peak in the X-ray diffraction (XRD) profile also suggests the consumption of FA^+ in the perovskite films after reverse-bias ageing (Supplementary Fig. 7).

We further used density functional theory (DFT) calculations to identify the FA^+ degradation pathways under reverse bias. FA^+ is believed to be chemically inert in the perovskite lattice. However, the potential contact with ITO at the buried interface (Supplementary Fig. 8) could trigger reactions of FA^+ and lead to the degradation of FA^+ as well as the whole perovskite lattice that loses A-site cations continuously. The molecular structure of FA^+ is such that its degradation could proceed through three kinds of elementary reactions: N-H cleavage, C-H cleavage and N-C cleavage (Fig. 1e). Our calculations show that N-H cleavage is the dominant reaction both thermodynamically and kinetically. The reaction from NH_2CHNH_2 (FA^+) to NHCHNH_2 is a highly exothermic

reaction (-0.81 eV) with an activation barrier of 0.93 eV, while the activation barriers for C-H cleavage and N-C cleavage are markedly higher (Fig. 1f and Supplementary Figs. 9–11). Furthermore, for the generated NHCHNH_2 , N-H cleavage is still the dominant reaction with an even lower activation barrier of 0.73 eV (Fig. 1f and Supplementary Figs. 12–14). Finally, the degradation reaction will terminate at the intermediate NHCHNH_2 , as all its possible elementary reactions have extremely high activation barriers (Fig. 1f and Supplementary Figs. 15–17). Thus, our calculations show that FA^+ will be consumed on the ITO surface by continuous N-H cleavage, with the first N-H cleavage step being the rate-determining step in the degradation of FA^+ .

The activation barrier of 0.93 eV is sufficiently high to slow the degradation of FA^+ under the operating conditions of perovskite solar cell devices at mild temperatures. However, under reverse bias, both the electric field and injected electrons could influence the potential energy diagram of the FA^+ degradation reaction by modifying

the electronic structures and corresponding bond strength of FA^+ . According to our calculations, although the electric field could lower the activation barriers moderately, the introduction of extra electrons changes the reaction kinetics radically: the activation barriers for both N–H and C–H cleavage become 0 eV in a variety of chemical environments (Supplementary Fig. 18), while the activation barrier for C–N cleavage is barely affected (Fig. 1g). These results indicate that the introduced electron could be selectively injected into the antibonding states of hydrogen-containing bonds and accelerate the degradation reaction greatly, implying the reduction of H^+ by the injected electron. Calculations showed a similar effect for the degradation of NHCHNH_2 (Supplementary Fig. 19) and NHCHNH (Supplementary Fig. 20), but to a lesser degree (the reactions will no longer have a zero activation barrier). However, the degradation of NHCHNH will not be terminated under conditions with accumulated carriers; the reverse bias could markedly accelerate the degradation reaction to a greater degree by changing the activation barrier from 0.93 eV to 0.40 eV and achieving degradation products beyond NHCHNH (Supplementary Fig. 21), which could be the main issue that influences the stability of PSCs.

Molecular-templated pre-assembly reinforces buried interface

To achieve high-performing SAM-based PSCs and modules with robust reverse-bias resistance, it is critical to suppress current leakage and continuous FA^+ degradation by isolating the perovskite from ITO. Therefore, enhancing the coverage on ITO through a more compact SAM layer is key to achieve perfect isolation. Two key factors affect SAM coverage. First, the state of aggregation of SAM molecules in the precursor solution significantly impacts their uniformity of distribution in wet film when deposited on the substrate²⁷. In addition, limited interactions between the anchoring phosphoric acid groups and the ITO substrate²⁸ allows SAM molecules to be easily affected by solvent flow dynamics before solvent evaporation, resulting in incomplete coverage. In this study, the discontinuous distribution of the SAM layer on ITO was confirmed by atomic force microscopy–infrared (AFM–IR) spectroscopy (Fig. 2h and Supplementary Fig. 22)²⁹. To enhance SAM coverage, we proposed a strategy that ensures the uniform pre-distribution of SAM molecules in the precursor solution and promotes the formation of clusters with stronger substrate interactions, thereby mitigating the effects of solvent flow dynamics. This was realized by introducing as a template poly[4-(9H-carbazol-9-yl)butyl]phosphonic acid] (Poly-4PACz), a polymer with a higher density of uniformly distributed anchoring groups³⁰ that bind with SAM and ITO, facilitating the deposition of the SAM. Pre-assembled SAM clusters could be formed in the solvent to promote the coverage and compactness of the SAM on the ITO substrate.

DFT calculations and molecular dynamics (MD) simulations were conducted to prove the pre-assembly effect introduced by Poly-4PACz both thermodynamically and kinetically. Atomic structural sampling showed that Poly-4PACz adopts an alternating antiparallel stacking motif (Supplementary Fig. 23), with repeating units comprising two SAM monomers whose phosphonic acid groups are oriented oppositely, resulting in an ordered distribution. The simulated atomic geometry of Poly-4PACz was further mixed with Ph-4PACz in dimethylsulfoxide (DMSO) as the initial state of MD simulations. After 70 ns, the ordered distribution pattern, similar to the DFT-simulated pattern, was preserved due to the rigidity of the antiparallel stacked carbazole groups. Furthermore, this ordered pattern could efficiently capture Ph-4PACz in solvent through a composite hydrogen-bonding network, that is, two hydrogen bonds could form simultaneously between the phosphoric acid groups of Poly-4PACz and Ph-4PACz (Fig. 2a and Supplementary Fig. 24) to achieve the pre-assembly of Ph-4PACz. Meanwhile, DMSO interacts only with the carbazole groups of Poly-4PACz and acts as a bystander in the pre-assembly process (Supplementary Figs. 24 and 25). Our simulations show that the strength of the interaction between Poly-4PACz and Ph-4PACz is

highly dependent on the coverage of Ph-4PACz on Poly-4PACz (Fig. 2b). Compared with full pre-assembly (that is, all phosphoric acid groups of Poly-4PACz are occupied by Ph-4PACz), half pre-assembly is more energetically favourable, in accord with the MD simulation, where nearly 45% of the phosphonic acid groups on the Poly-4PACz template are engaged in assembly with Ph-4PACz in DMSO after 70 ns (Supplementary Fig. 24d). Meanwhile, the unoccupied phosphonic acid groups can interact with the ITO substrate, further anchoring the ordered pre-assembled Ph-4PACz on the ITO surface. Based on the half assembly of Ph-4PACz on Poly-4PACz, the interactions between Poly-4PACz and Ph-4PACz came to dominate the self-interactions of Ph-4PACz and Poly-4PACz (Supplementary Figs. 26 and 27). Furthermore, our calculations demonstrate that the pre-assembly process is enthalpy-dominated, although configurational entropy also contributes (Supplementary Fig. 28 and Supplementary Note 3). Therefore, Poly-4PACz serves as a template for the pre-assembly of the small Ph-4PACz molecule in the precursor solution, a process we refer to as the polycarbazole-templated pre-assembly of the SAM (PTP-SAM).

To demonstrate this experimentally, we employed solution-phase Fourier transform infrared (FTIR) spectroscopy to probe the chemical environment of the SAM, Poly-4PACz and PTP-SAM in DMSO. Upon introducing Poly-4PACz, the P–O–H stretching peak shifted from 981 cm^{-1} to 984 cm^{-1} , indicating the formation of a new hydrogen-bonding-induced pre-assembly pattern in the precursor solution (Fig. 2d). This shift was further corroborated by DFT calculations: the P–O–H stretching peak in SAM was predicted at 988 cm^{-1} and at 991 cm^{-1} in PTP-SAM (Supplementary Fig. 29). We also performed diffusion-ordered spectroscopy in $[\text{D}_6]\text{DMSO}$ (Supplementary Fig. 30 and Supplementary Note 4). SAM showed the highest $\log[D\text{ (m}^2\text{ s}^{-1})]$ (-9.4) (where D is the translational diffusion coefficient), with a uniform distribution, Poly-4PACz the lowest (-10.2), with heterogeneous segments, and PTP-SAM an intermediate value (around -9.8), with a more uniform distribution, indicative of homogeneous clusters formed through the pre-assembly process. In addition, ^{31}P NMR spectroscopy was conducted to probe the phosphonic acid groups of the SAM, Poly-4PACz and PTP-SAM (Fig. 2e). PTP-SAM showed a downfield shift and pronounced peak broadening, indicating a modified phosphorus environment arising from newly formed hydrogen bonds and additional SAM–Poly-4PACz associations. We also used in-solution small-angle X-ray scattering (SAXS) to probe the solution-phase microstructures arising from the interactions of the SAM, Poly-4PACz and PTP-SAM with the solvent DMSO (Supplementary Fig. 31 and Supplementary Note 5). A positive low- q (where q is the scattering vector) deviation between the measured PTP-SAM curve and predicted non-interacting baseline from SAM and Poly-4PACz solutions indicates intercomponent attraction and the emergence of larger-scale correlations beyond simple superposition^{31,32}. Although Poly-4PACz tends to aggregate significantly in DMSO solution due to strong intermolecular interactions between multiple phosphonic acid groups (colloidal size of $\sim 60\text{ nm}$, measured by dynamic light scattering), the final PTP-SAM has a smaller colloidal size (0.8 nm) compared with the SAM solution (1.05 nm), facilitating uniform deposition (Supplementary Fig. 32). These results are consistent with our hypothesis that the SAM undergoes pre-assembly on Poly-4PACz templates through hydrogen bonding.

To demonstrate the persistence of pre-assembly in solid films, we employed synchrotron-based grazing-incidence SAXS (GISAXS) to probe the ordered microstructure in final films (Fig. 2f,g and Supplementary Fig. 33). In the vertical integration profiles, the SAM film exhibits a broad feature centred at low q (-0.12 – 0.15 nm^{-1}), indicative of relatively large, polydisperse domains and pronounced mesoscale heterogeneity, while the Poly-4PACz film shows a sharper peak at a higher q (-0.24 – 0.26 nm^{-1}), consistent with more strongly aggregated polymer clusters. By contrast, PTP-SAM shows a narrow feature at an intermediate q (-0.20 – 0.22 nm^{-1}), together with a suppressed low- q upturn, evidencing a more uniform (that is, narrow

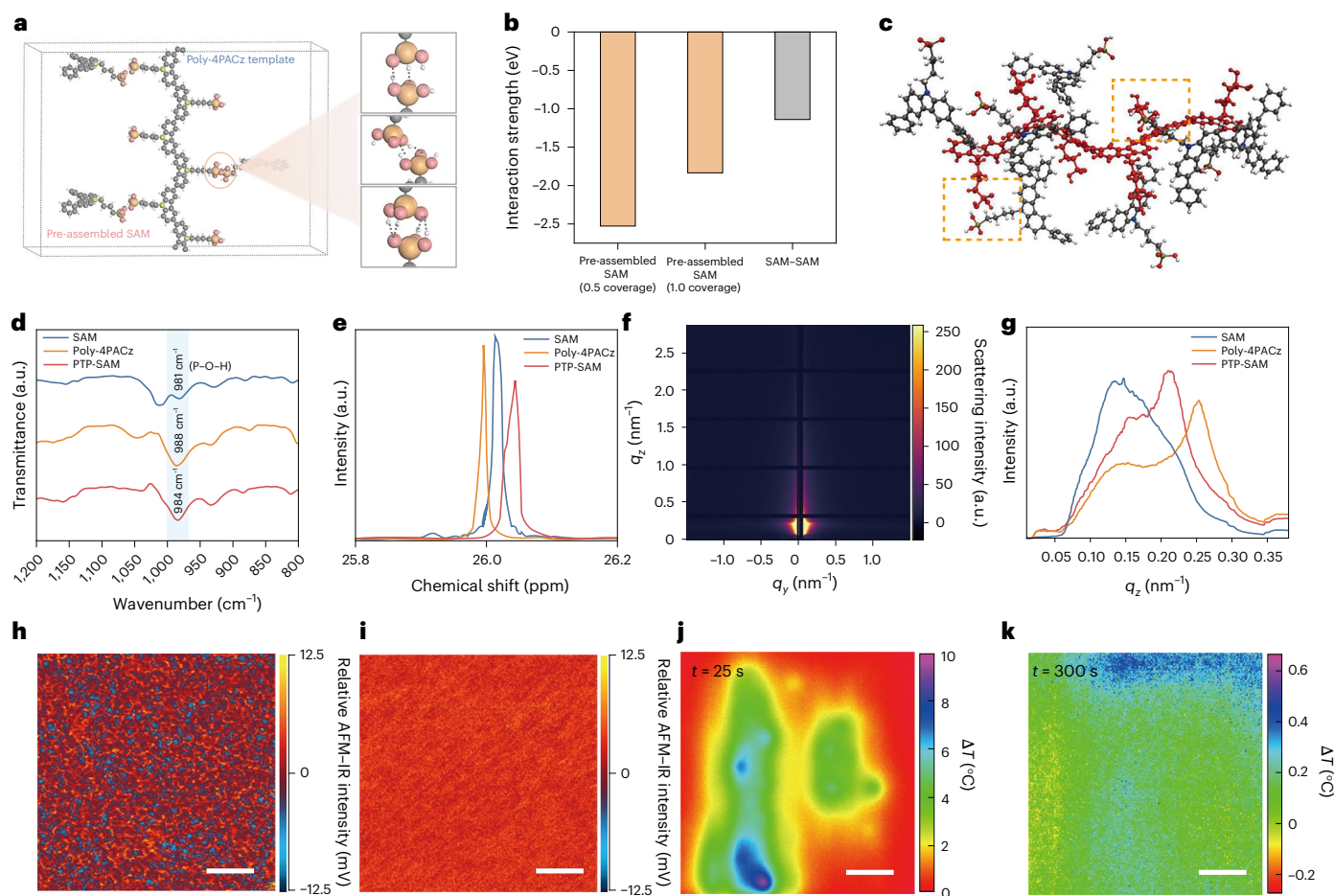


Fig. 2 | Molecular-templated pre-assembly reinforces the buried interface.

a, Diagram of the assembled SAM templated by Poly-4PACz. Insets: modes of interaction between the SAM and Poly-4PACz. Orange spheres denote P atoms, red spheres denote O atoms, green spheres denote N atoms, grey spheres denote C atoms and white spheres denote H atoms. **b**, Strength of the interaction between Poly-4PACz and SAM, with the SAM–SAM interaction shown as reference. **c**, Snapshot of an MD simulation showing an assembled pattern with one Poly-4PACz template associating with two SAM molecules (the red molecule is Poly-4PACz, the black molecules are Ph-4PACz and the orange dashed boxes highlight the phosphate groups involved in the interaction between

Ph-4PACz and Poly-4PACz). **d**, FTIR spectra of the SAM, Poly-4PACz and PTP-SAM in DMSO solution. The blue shaded bar indicates the P–O–H vibrational peak. a.u., arbitrary unit. **e**, ^{31}P NMR spectra of the SAM, Poly-4PACz and PTP-SAM in $[\text{D}_6]\text{DMSO}$. **f**, Two-dimensional GISAXS image of the PTP-SAM film. **g**, GISAXS vertical integration profile (q_z) of the SAM, Poly-4PACz and PTP-SAM. **h, i**, AFM–IR intensity maps of the SAM (**h**) and PTP-SAM (**i**) films on ITO substrate. Scale bars, 200 nm. In **i**, the IR signal was captured at $1,495\text{ cm}^{-1}$ (ref. 29). **j, k**, Lock-in thermography images of SAM-based (**j**) and PTP-SAM-based (**k**) perovskite solar modules after $-V_{\text{MPP}}$ bias in air for 25 and 300 s, respectively. Scale bars, 1 cm.

distribution), well-defined nanostructure that is not a simple superposition of the SAM and Poly-4PACz. In addition, we performed an MD simulation of PTP-SAM with the initial state extracted from solution after pre-assembly (Supplementary Fig. 34). The resulting film shows no macroscopic phase separation or layering. These results indicate that the PTP-SAM forms a new mixed packing motif, supporting the persistence of the pre-assembly structure in the final film. Furthermore, AFM–IR mapping, XPS and Kelvin probe force microscopy showed that the coverage uniformity of the SAM layer is significantly improved (Fig. 2i, Supplementary Figs. 35–37 and Supplementary Note 6), while the crystallinity, composition and buried interface properties of the overlying perovskite remain unaffected (Supplementary Fig. 38). Also, to exclude non-specific effects from polymer additives on solution rheology or film formation, poly(9-butyl-9H-carbazole) (PolyCz-C4H9), a polymer with a similar structure but without the terminal phosphonic acid groups, was synthesized and applied in the same way as Poly-4PACz. Both Poly-4PACz and SAM/PolyCz-C4H9 films showed poor coverage (Supplementary Figs. 39 and 40 and Supplementary Note 7), confirming the importance of the pre-assembly process. Furthermore, we employed lock-in thermography to assess the

enhanced robustness and compactness of the buried interface over a large area. SAM- and PTP-SAM-based solar modules with an area of 20 cm^2 were biased at the negative voltage at the maximum power point ($-V_{\text{MPP}}$, where $V_{\text{MPP}} = 8.1\text{ V}$) in air to monitor increases in temperature under reverse bias. In the SAM-based device, hot spots appeared within seconds of reverse biasing, indicating reverse current flow and localized Joule heating (Fig. 2j and Supplementary Fig. 41a–c). The average temperature in some regions increased by 5°C within 25 s (Supplementary Fig. 42a,b). In contrast, no hot spots were observed in the PTP-SAM-based device, even after applying a bias for 5 min (Fig. 2k and Supplementary Fig. 41d–f), and the temperature across the whole device remained largely unchanged (Supplementary Fig. 42c,d).

Small-area PSC performance

A cross-sectional scanning electron microscopy (SEM) image of the optimized PTP-SAM device is shown in Fig. 3a. Employing PTP-SAM to minimize current leakage at the buried interface led to an increase in average PCE from 25.7% to 26.1% (Supplementary Fig. 43), primarily due to improvements in the fill factor (FF) and open-circuit voltage (V_{oc}), which can be attributed to enhanced shunt resistance, favourable

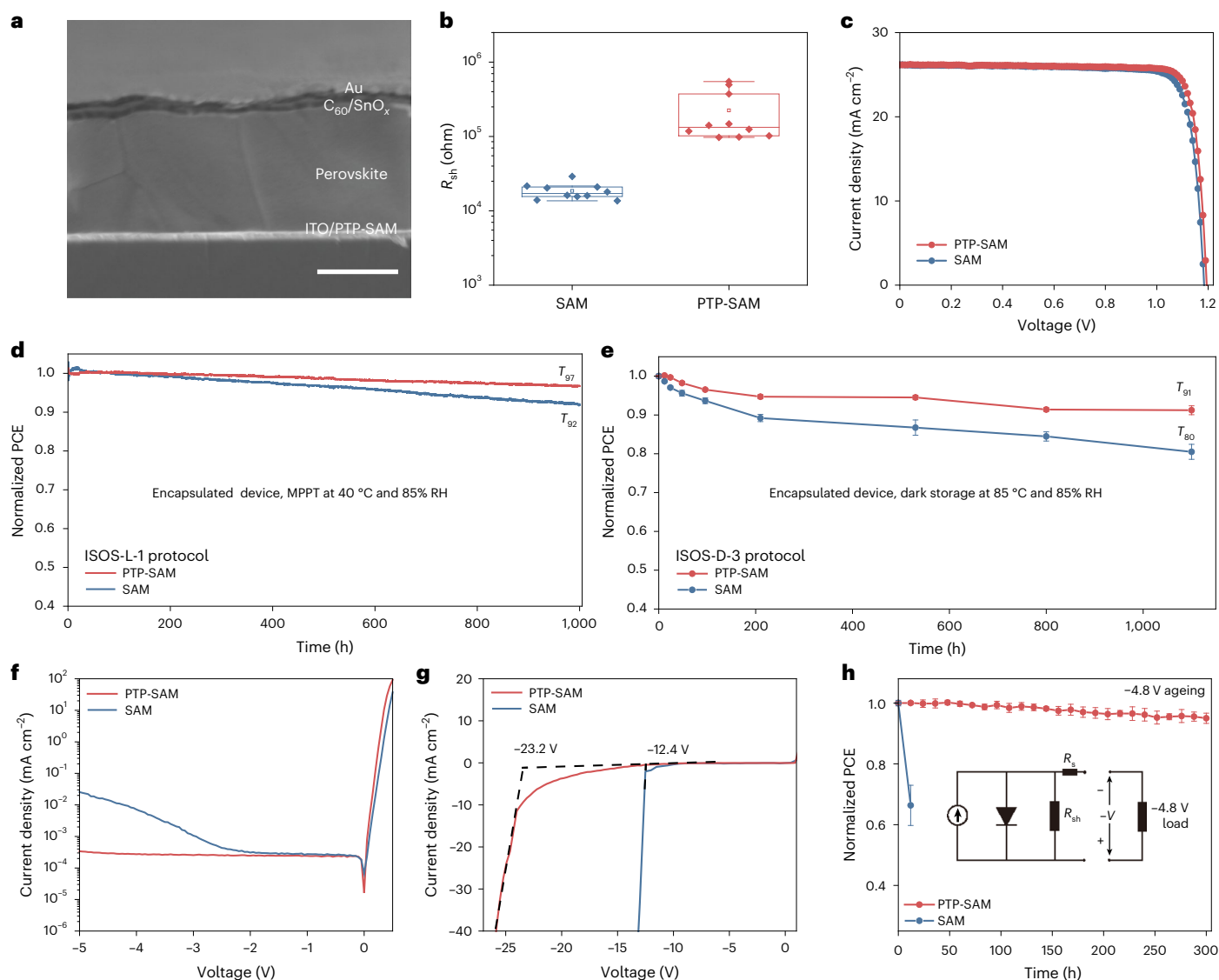


Fig. 3 | Efficiency and long-term stability of PSCs. **a**, Cross-sectional SEM image of a PSC with the architecture of ITO/PTP-SAM/Cs_{0.1}FA_{0.9}PbI₃/C₆₀/SnO_x/Au. Scale bar, 500 nm. **b**, Statistical shunt resistance (R_{sh}) of PSCs with different buried interface. The data are presented as the median (line) and mean (open square), calculated for ten devices (diamonds) under each condition; the lower and upper box lines represent the 25–75% quartiles and the whiskers denote the 1.5× interquartile range. **c**, J – V scans (from 1.2 V to 0 V) of PSCs (0.05 cm²) with different buried interfaces. **d**, Continuous MPPT of encapsulated PSCs with SAM and PTP-SAM under the ISOS-L-1 protocol (initial efficiency: 25.50% for SAM and 26.12% for PTP-SAM; T_{97} and T_{92} are defined as the time required for the device performance to decay to 97% and 92%, respectively, of its initial value).

e, Thermal stability of PSCs with SAM and PTP-SAM under the ISOS-D-3 protocol (initial average efficiency: 24.57% for SAM and 24.84% for PTP-SAM; T_{91} and T_{80} are defined as the time required for the device performance to decay to 97% and 92%, respectively, of its initial value). The data are presented as the mean ± s.d. from six devices. **f**, Dark J – V curves of PSCs with SAM and PTP-SAM. **g**, Breakdown voltage of PSCs with SAM and PTP-SAM (the dashed lines are drawn as guides to the eye). **h**, PCE evolution of the target PSCs under ageing at –4.8 V in the dark (25 ± 5 °C, nitrogen atmosphere, initial average efficiency: 24.19% for SAM and 25.22% for PTP-SAM). The data are presented as the mean ± s.d. from five devices. Inset: equivalent circuit diagram of the ageing condition. R_s , series resistance.

band alignment and reduced recombination at the buried interface (Fig. 3b and Supplementary Figs. 44 and 45). The best-performing target device with a PTP-SAM layer showed a maximum PCE of 26.3% with a V_{oc} of 1.185 V, FF of 85.1% and short-circuit current density (J_{sc}) of 26.11 mA cm^{–2}, while a control device with a SAM layer showed a maximum PCE of 25.9% with a V_{oc} of 1.177 V, FF of 84.2% and J_{sc} of 26.16 mA cm^{–2} (Fig. 3c). The integrated J_{sc} of the target device was calculated to be 25.7 mA cm^{–2} from the external quantum efficiency (EQE), which is consistent with the J – V curves (Supplementary Fig. 46). Both the control and target devices exhibited excellent operational stability, retaining over 90% of their initial PCE after 1,000 h 1-sun MPPT (following the International Summit on Organic Solar Cell Stability protocol for light-soaking stability testing under continuous 1-sun illumination at

ambient temperature and ambient atmosphere (ISOS-L-1), Fig. 3d). The slightly enhanced operational stability of the target device is attributed to the improved robustness at the buried interface that minimizes ion migration. Regarding damp-heat stability (International Summit on Organic Solar Cell Stability protocol for dark-storage stability testing under 85 °C and 85% RH (ISOS-D-3), Fig. 3e), the target device retained 91% of its initial PCE, while the control device kept only 80% after ageing for 1,100 h, indicating that the PTP-SAM exhibits better thermal stability through a stronger interaction with the ITO substrate.

Moreover, the target devices showed markedly improved reverse-bias stability compared with their SAM-based counterparts due to reduced current leakage and suppressed FA⁺ degradation at the buried interface (Supplementary Fig. 47). The target devices

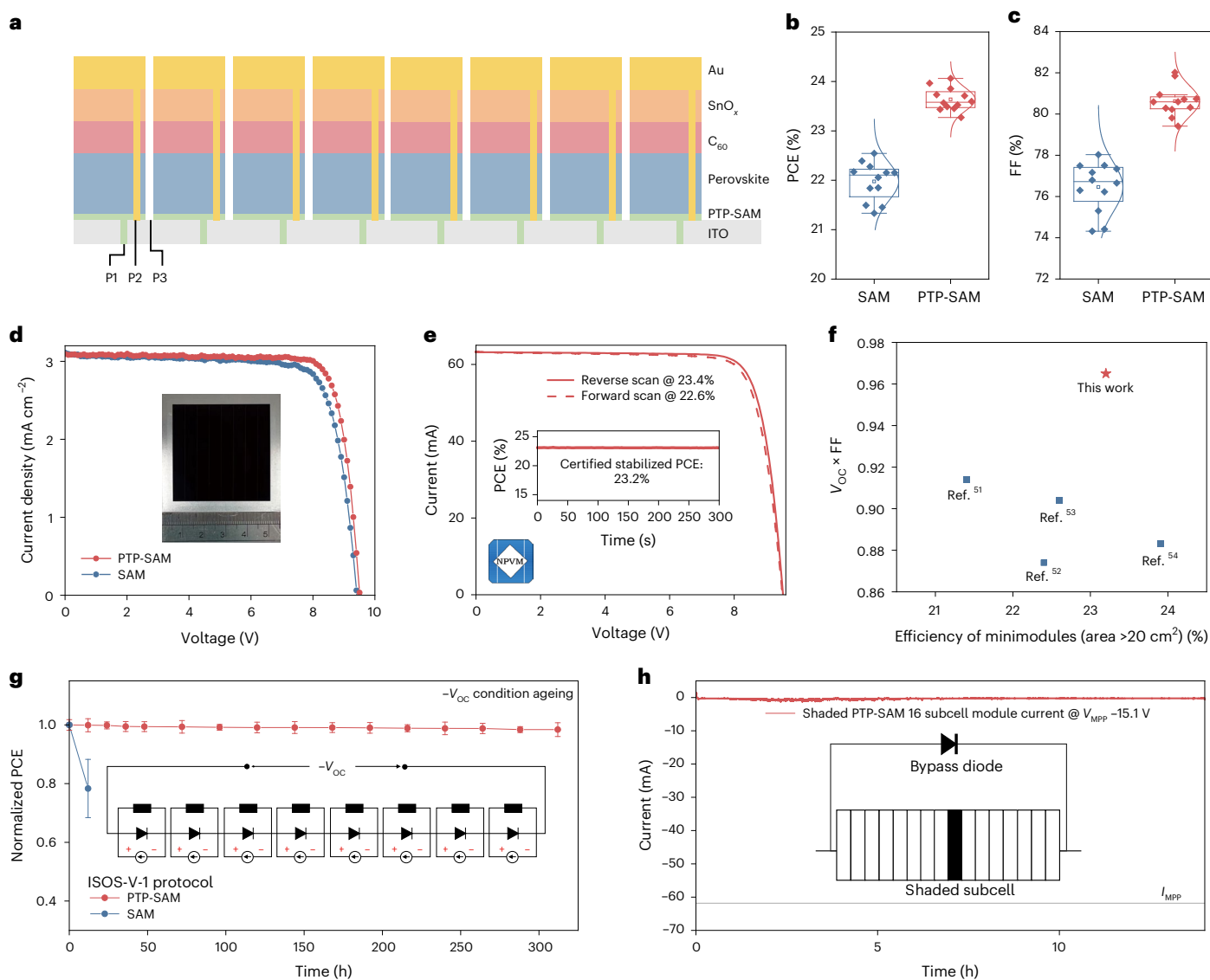


Fig. 4 | Efficiency and long-term stability of PSMs. **a**, Diagram of a perovskite solar minimodule with eight subcells. P1, P2 and P3 represent the laser scribing steps for bottom electrode isolation, interconnection and top electrode isolation, respectively. **b,c**, Statistical PCE (**b**) and FF (**c**) of 20.25-cm² PSMs with different buried interfaces. The data are presented as the median (line) and mean (open square), calculated for ten devices (diamonds) under each condition; the lower and upper box lines represent the 25–75% quartiles and the whiskers denote the 1.5× interquartile range. **d**, J–V measurement of 20.25-cm² PSMs with different buried interfaces. Inset: photograph of the target PSM with a mask. **e**, J–V measurement of a 20.25-cm² target PSM by NPVM. Inset: stabilized output

of a 20.25-cm² device by NPVM. **f**, Summary of certified efficiencies and $V_{OC} \times FF$ values for perovskite solar minimodules from this work and refs. 34, 51–54. **g**, PCE evolution of an unencapsulated control and target minimodules aged under $-V_{OC}$ at -9.5 V and 25 ± 5 °C in the dark under a nitrogen atmosphere (initial average efficiency: $21.83\% \pm 5\%$ for SAM and $23.16\% \pm 5\%$ for PTP-SAM). The data are presented as the mean $\pm$ s.d. from five devices. Inset: equivalent circuit diagram of the ageing condition. **h**, Current evolution of unencapsulated target minimodules with 16 subcells aged under 1-sun illumination at V_{MPP} operating conditions. One subcell was shaded with black tape (55 ± 5 °C, nitrogen atmosphere). Inset: schematic illustrating a single bypass diode protecting 16 subcells.

maintained a consistently low reverse current when increasing the reverse bias up to -5.0 V, indicating a strong electron-blocking capability at the buried interface (Fig. 3f and Supplementary Fig. 48), whereas the SAM-based devices showed a sharp increase in current when the reverse bias was increased beyond -2.5 V. Furthermore, the V_b of the target device improved to -23.2 V (Fig. 3g), the highest value observed so far for inverted PSCs with a metal top electrode^{14,15}. We evaluated the long-term stability of devices under a continuous reverse bias of -4.8 V (see Methods for details), approximately four times the ISOS-V-1 specification³³, enabling accelerated testing. The control devices retained only 66% of their initial PCE after 12 h, while the target devices maintained 95% of their initial PCE after ageing for 300 h (Fig. 3h). Considering multifactorial stress in real-world operation,

we initiated combined stressor tests ($-V_{OC}$ reverse bias plus 85 °C with an environmental relative humidity (RH) of 88%) to evaluate the stability under more practical conditions (Supplementary Fig. 49). The PTP-SAM device showed a T_{80} (T_{80} is defined as the time required for the device performance to decay to 80% of its initial value) lifetime of 9 h, whereas the SAM device degraded more rapidly, dropping to 65% of its original performance after only 1 h and breaking down completely after 2 h. These results, consistent with our single stressor tests, further demonstrate the robustness of the PTP-SAM strategy.

Performance of large-area perovskite solar modules

As PSCs are scaled up, increased current leakage at the buried interface due to direct perovskite–electrode contact amplifies performance loss

and reverse-bias degradation. Therefore, perovskite solar modules (PSMs) with an aperture area of 20.25 cm² were fabricated to demonstrate the effect of PTP-SAM over a larger area. Eight subcells were connected in series via laser scribing, achieving a geometric fill factor of 97.1% (Fig. 4a and Supplementary Fig. 50). The average FF improved from 76.4% (control PSMs) to 80.5% (target PSMs), resulting in an impressive average PCE exceeding 23.5% for the target PSMs (Fig. 4b,c and Supplementary Fig. 51). This result further confirms that current leakage over a larger area is effectively suppressed by the PTP-SAM. The best-performing target PSM achieved a PCE of 24.0%, with a V_{oc} of 9.50 V, FF of 82.0% and J_{sc} of 3.08 mA cm⁻². For comparison, the control PSM exhibited a V_{oc} of 9.41 V and FF of 77.5%, corresponding to a PCE of 22.5% (Fig. 4d). One target PSM was sent to the National Photovoltaic Industry Metrology Test Center (NPVM) for certification, which confirmed a reverse-scan PCE of 23.4% and a stabilized MPPT efficiency of 23.2% with the highest $V_{oc} \times FF$ (Fig. 4e and Supplementary Figs. 52 and 53). This certified value marks the first reported steady-state PCE to surpass 23% at the module level and has been included in the solar cell efficiency tables in ref. 34 (Fig. 4f and Supplementary Table 1).

To evaluate the reverse-bias stability at the minimodule scale, we applied a constant reverse voltage of $-V_{oc}$ (-9.5 V) to both control and target PSMs and monitored the evolution of PCE over time following the ISOS-V-1 protocol (see Supplementary Fig. 54 for a photograph of the set-up). The control PSMs exhibited rapid degradation, retaining only 78% of their initial PCE after ageing for 12 h, whereas the target PSMs maintained 98% of their original performance after 312 h, demonstrating excellent robustness under reverse-bias stress (Fig. 4g, Supplementary Fig. 55 and Supplementary Note 8). For comparison, previously reported PSMs retained only 84% of their initial performance after just 1 h under $-V_{MPP}$ bias³⁵. To further validate the effectiveness of our strategy, we subjected a single subcell (out of eight) to partial shading and aged the minimodules under 1-sun illumination at 55 ± 5 °C in a short-circuit state (see Supplementary Figs. 56 and 57 for photographs of the set-up and equivalent circuit diagram). Under these conditions, the shaded subcell experienced the highest reverse stress with a reverse voltage of $-7 \times V_{cell,sc}$ (the voltage of each subcell at short-circuit condition) to accelerate ageing (Supplementary Fig. 57). The control device degraded to below 80% of its initial PCE within just 1 h, whereas the target PSM retained 90% of its original PCE after ageing for 8 h (Supplementary Fig. 58).

We simulated a more realistic scenario in which a subcell was shaded while the module operated at its maximum power point. If the subcell underwent breakdown, the output current of the entire module would rapidly rise to the module current at maximum power point (I_{MPP}) (refs. 15,36). We monitored the current of a 16-subcell PTP-SAM module with one subcell shaded under 1-sun illumination at V_{MPP} (Supplementary Fig. 59). The current output remained at a low level (average current $I_{avg} = -456.1$ μ A) for 14 h, without current leakage to the milliampere level or sudden breakdown to I_{MPP} . Meanwhile, the module retained 97% of its initial performance (Supplementary Fig. 60). This indicates that only one single bypass diode can effectively protect a perovskite module with at least 16 subcells under real operating conditions (Fig. 4h).

Conclusion

We have identified discontinuous SAM coverage and ITO-induced electrochemical deprotonation of FA⁺ at the buried interface as the primary origins of reverse-bias instability in high-efficiency inverted PSMs. Guided by this insight, we developed a polycarbazole-templated pre-assembly strategy that drives homogeneous clustering in solution and yields a dense, uniform SAM layer. This interface engineering enhances both performance and robustness, enabling 20.25-cm² minimodules with a certified steady-state efficiency of 23.2% while suppressing reverse-bias degradation under accelerated ISOS protocols. Our findings highlight the electrochemistry at the buried interface as

a critical reliability bottleneck and provide guidance for integrating ultrathin SAM-based contacts in scalable perovskite photovoltaics.

Methods

Materials

Caesium iodide (CsI, 99.99%), lead chloride (PbCl₂), dimethylformamide (DMF, anhydrous) and 1-methyl-2-pyrrolidinone (NMP, anhydrous) were purchased from Sigma-Aldrich. Formamidinium iodide (FAI) and methylammonium chloride (MACl) were purchased from Greatcell Solar Materials. Anhydrous lead iodide (PbI₂) was purchased from Tokyo Chemical Industry, gold was purchased from Alfa Aesar and C₆₀ was purchased from Lumtec. Poly-4PACz (weight-average molecular mass (M_w) = 2,000–3,000) was purchased from Borun New Material Company. Ph-4PACz was synthesized according to our previous report¹⁸.

Device fabrication

For small-area devices, patterned ITO glass substrates were wiped with soap water, then ultrasonically washed with deionized water, acetone and isopropanol for 30 min each. After ultraviolet–ozone treatment for 15 min or 450 W air plasma treatment for 5 s, the ITO substrates were transferred to a nitrogen-filled glove box. Then, Ph-4PACz (1 mg ml⁻¹ in DMSO) or Ph-4PACz–Poly-4PACz (mass ratio = 9:1, 1 mg ml⁻¹ in DMSO) was spin-coated onto the ITO substrates at 5,000 r.p.m. for 30 s and heated at 100 °C for 10 min. A 1.8 M perovskite (Cs_{0.1}FA_{0.9}PbI₃) stock solution was prepared by dissolving 52.0 mg CsI, 309.5 mg FAI, 1,014.2 mg PbI₂ and 13.5 mg MACl in 1.11 ml DMF–NMP (v/v = 960:150). The 1.8 M perovskite precursor solution was shaken overnight to ensure that the components were fully dissolved and then used to prepare perovskite films. For the spin-coating process, the substrate was spun at 5,000 r.p.m. for 40 s with an acceleration of 5,000 r.p.m. per second and N₂ gas was blown on top of the spinning substrate after 20 s. The perovskite films were pre-annealed at 70 °C for 2 min in a N₂-filled glove box and then annealed at 150 °C for 10 min in an RH of 30%. For passivation, 0.5 mg ml⁻¹ 1,3-propanediammonium iodide (PDAI) in isopropyl alcohol (IPA) solution was spin-coated onto the perovskite film and then heated at 100 °C for 5 min. Then, C₆₀ (50 nm) was deposited via evaporation and SnO_x (25 nm) was deposited by atomic layer deposition. Finally, Au (100 nm) was deposited to complete the device fabrication. The devices were masked with metal aperture masks (0.05 cm²) during J – V measurement.

PSMs with eight subcells connected in series were fabricated on ITO glass with a size of 5.5 cm × 5.5 cm; the width of one subcell was 5.625 mm. The series interconnection of the module was realized by P1, P2 and P3 lines, which were patterned using a laser scribing system with a wavelength of 1,064 nm for P1 and 532 nm for P2 and P3 with a power of 20 W. The ITO substrate was pre-patterned for P1 (30 μ m wide) at 60% laser power and a speed of 200 mm s⁻¹ with a frequency of 30 kHz and pulse width of 30 ns. The HTL, perovskite layer, passivation layer and C₆₀/SnO_x layers were then prepared in the same way as for the small-area devices. The P2 lines (33 μ m wide) were patterned with an average laser power of 30% at a speed of 200 mm s⁻¹ and frequency of 30 kHz. After the deposition of 100 nm Au, the P3 lines (37 μ m wide) were patterned with an average laser power of 28% at a speed of 200 mm s⁻¹ and frequency of 30 kHz. The devices were masked with metal aperture masks (20.25 cm²) during J – V measurement. The distance between the P1 and P3 lines was about 160 μ m and the geometric FF was around 97.1%.

Device characterization

J – V measurements of the devices were performed with a Keithley 2400 source meter under simulated 1-sun AM1.5G illumination (100 mW cm⁻²) using an ABET Technologies Sun 2000 solar simulator. The measurements were performed in an N₂-filled glove box. The illumination area of the devices was defined by shadow masks with areas of 0.05 and 20.25 cm². MPPT measurements of the encapsulated

devices were recorded using an MPP Tracking-4B system (Shenzhen Lancheng Technology) and an AM1.5G spectrum simulated with a light-emitting diode (LED). Measurements were carried out under ambient conditions at -45°C with an RH of 85% (ISOS-L-1). Damp–heat ageing stability tests were conducted by putting the encapsulated solar cells in an environment chamber at 85°C with an RH of 85%, periodically taking measurements (ISOS-D-3). EQE measurements were conducted using a Bentham PVE300-IVT system. The LED laser intensity was calibrated using built-in silicon and germanium diodes before measurements. For high-resolution EQE measurements, the light was chopped at 137 Hz and coupled to a Bentham monochromator. The resulting monochromatic light was focused on the perovskite solar cell and its current under short-circuit conditions was fed to a current pre-amplifier (Stanford SR 570) before analysis with a lock-in amplifier (Stanford SR830 DSP). The time constant of the lock-in amplifier was chosen to be 1 s and the amplification of the pre-amplifier was increased to resolve low photocurrents. EQEs were calculated by dividing the photocurrent of the cell by the flux of incoming photons, which was measured using a calibrated Si photodiode. All EQE measurements were performed under ambient conditions at 25°C with a RH of around 60% and without encapsulation.

For lock-in thermography, in situ thermal imaging was conducted using a FLIR A655sc infrared camera, primarily operating at frame rates between 1 and 5 Hz, depending on the measurement conditions. A programmable DC source (SourceMeter) was used to apply controlled forward and reverse biases to the devices. Reverse-bias experiments were performed predominantly under -8.1 V . During measurements, no external illumination was applied and the experiments were carried out under ambient air conditions with temperatures between 22 and 24°C . Real-time temperature maps were recorded and analysed. The thermal images were processed using FLIR ResearchIR and MATLAB to extract localized temperature changes. This methodology enabled the visualization and study of localized hot spot formation, evolution and disappearance under electrical stress, particularly under reverse-bias conditions.

Reverse-bias stability test protocols

As shown in Fig. 3h, each data point was collected by I – V scanning after the following treatment: (1) biasing the small-area solar cell at -4.8 V in a dark, nitrogen-filled glove box for 12 h (ISOS-V-1), (2) recovery in the dark for 10 h and (3) light soaking under 1-sun illumination for 30 min before measurement. The pre- and post-recovery data, as well as the evolution of the reverse current for 12 h in one cycle, are presented in Supplementary Figs. 61 and 62.

As shown in Fig. 4g, each data point was collected by I – V scanning after the following treatment: (1) biasing the PSM at -9.5 V in a dark, nitrogen-filled glove box for 12 or 24 h for (ISOS-V-1), (2) recovery in the dark for 12 h and (3) light soaking under 1-sun illumination for 30 min before measurement.

As shown in Supplementary Fig. 58, each data point was collected by I – V scanning after the following treatment: (1) the shaded PSM was placed in short-circuit conditions under 1-sun illumination at $55 \pm 5^{\circ}\text{C}$ inside a nitrogen-filled glove box for 1 h (ISOS-V-2) and (2) the shaded mask was removed and recovered in the dark for 1 h before measurement.

Materials characterization

SEM images were taken with a Regulus SU8200 microscope (Hitachi) at an accelerating voltage of 3 kV in secondary electron (SE) mode. XRD was performed on a Malvern Panalytical X-ray diffractometer with Cu K α radiation ($\lambda = 0.1542\text{ nm}$). AFM–IR spectroscopy was conducted with a Park NX20 spectrometer. The spatial distribution of the AFM–IR spectra was characterized using a nanoIR2 system (Anasys Instruments) by detecting IR signals at $1,494\text{ cm}^{-1}$ (the aromatic C=C stretching vibration in carbazole). Both height and chemical images

were obtained in contact mode using a contact mode AFM probe for nanoIR2 in an ambient atmosphere. Solution FTIR spectroscopy was conducted on a NICOLET iS50 FTIR spectrometer in attenuated total reflection mode. XPS was performed with a Kratos Axis Ultra XPS system. Samples were stored under high-vacuum conditions overnight to minimize the influence of H_2O and other particles adsorbed on the surface. The original data were processed using the XPSPEAK41 software. ^1H NMR spectra were recorded on a Bruker spectrometer at 400 MHz. Samples were prepared by adding 100 μl [D_6]DMSO (recycled after each dissolution step) to each of ten aged devices following the removal of the electrode, SnO_2 and C_{60} layers to dissolve the perovskite films. The resulting [D_6]DMSO solutions were then combined and ^1H NMR spectra were recorded.

SAXS

SAXS and GISAXS experiments were performed at the NCD-SWEET beamline of the ALBA Synchrotron Light Source, Spain. X-rays were produced by an in-vacuum undulator with a period of 21 mm and monochromatized by a Si(111) channel-cut monochromator. Beam collimation was carried out using a stack of beryllium compound refractive lenses, delivering a spot size of approximately $120 \times 150\text{ }\mu\text{m}^2$ (vertical $\times$ horizontal, full-width half-maximum) at the sample position. The photon energy was set to 12.4 keV ($\lambda = 0.9998\text{ }\text{\AA}$). The scattered X-rays were collected with a Dectris Pilatus3 S 1M detector from Dectris(R), positioned at a sample-to-detector distance of 3,568 mm for SAXS and 3,546 mm for GISAXS. The detector geometry (distance, beam centre and tilt), and hence the reciprocal space, was calibrated using silver behenate from the National Institute of Standards and Technology (NIST) as a standard. For SAXS experiments, samples were sealed in borosilicate capillaries with a wall thickness of 10 μm and an outer diameter of 2 mm (Hilgenberg) and mounted perpendicular to the incident beam. GISAXS measurements were performed in a grazing-incidence geometry, scanning the incident angle close to the critical angle of the total external reflection to ensure surface sensitivity. SAXS data were reduced to one-dimensional profiles using pyFAI (ref. 37), while the GISAXS data were processed using Python home-built routines.

Computational details

Under the framework of the projector augmented wave method^{38,39}, all spin-polarized first-principles calculations were performed with the Vienna Ab initio Simulation Package (version 5.4.4)³⁹. The Perdew–Burke–Ernzerhof version of the generalized gradient approximation was used to depict the exchange–correlation effect between electron–electron interactions⁴⁰. The valence wave functions were expanded by plane wave using a cut-off energy of 400 eV, while all structures were optimized until the force on each atom was less than $0.02\text{ eV }\text{\AA}^{-1}$. The DFT–D3 (DFT–D3 is a semi-empirical dispersion correction method (Grimme’s third-generation) added to density DFT to account for long-range van der Waals interactions by introducing pairwise atomic dispersion terms to the total energy) method was applied for van der Waals correction⁴¹. A 2×2 (111) ITO surface was cleaved from optimal bulk crystal structures with a 15 $\text{\AA}$ vacuum layer, which was used as the surface model of ITO. The Brillouin zone was sampled using a $3 \times 3 \times 1$ k -point mesh for geometry optimization. To calculate the adsorption energy of different intermediates of FA degradation, the optimized molecules were further introduced onto the surface model of ITO. The adsorption energies (ΔE_{ads}) of different intermediates were calculated according to the following equation ($E_{\text{surface}+\text{CN}_2\text{H}_x}$ is the total energy of the surface with the adsorbed CN_2H_x intermediate; $E_{\text{surface}+\text{FA}}$ is the total energy of the surface with adsorbed formamidine (FA); E_{H_2} is the total energy of an isolated H_2 molecule; x is the number of hydrogen atoms removed from FA during the deprotonation process):

$$\Delta E_{\text{ads}} = E_{\text{surface}+\text{CN}_2\text{H}_x} + (2.5 - 0.5x)E_{\text{H}_2(\text{g})} - E_{\text{surface}+\text{FA}} \quad (1)$$

Moreover, all transition states of elementary reactions were searched using the climbing-image nudged elastic band method⁴² and further improved by the dimer method⁴³. The activation barrier (ΔE_a) was calculated according to the following equation (E_{TS} is energy of transition state and E_{IS} is energy of initial state):

$$\Delta E_a = E_{TS} - E_{IS} \quad (2)$$

The neutral molecules of the SAM (Ph-4PACz) were optimized in a $25 \text{ \AA} \times 25 \text{ \AA} \times 25 \text{ \AA}$ unit cell. Poly-4PACz was optimized in a $15 \text{ \AA} \times 30 \text{ \AA} \times 30 \text{ \AA}$ unit cell to screen the self-interactions of Poly-4PACz. To further accommodate the pre-assembled SAM, the unit cell used for pre-assembly calculations was expanded to $15 \text{ \AA} \times 30 \text{ \AA} \times 60 \text{ \AA}$. The Brillouin zone was sampled using a $3 \times 2 \times 1$ k -point mesh for geometry optimization.

The configuration entropy ($\Delta S_{\text{configuration}}$) of the pre-assembly process was calculated according to the ternary Flory–Huggins model:

$$\Delta S_{\text{configuration}} = -k_B (n_{\text{SAM}} \ln \phi_{\text{SAM}} + n_{\text{Poly}} \ln \phi_{\text{Poly}} + n_{\text{DMSO}} \ln \phi_{\text{DMSO}}) \quad (3)$$

where k_B is the Boltzmann constant, n_{SAM} , n_{Poly} and n_{DMSO} are the number of moles of SAM (Ph-4PACz), Poly-4PACz and DMSO, respectively, and ϕ_{SAM} , ϕ_{Poly} and ϕ_{DMSO} are volume fractions of the SAM, Poly-4PACz and DMSO, which were calculated according to the following equations:

$$\phi_{\text{SAM}} = \frac{n_{\text{SAM}}}{n_{\text{SAM}} + Nn_{\text{Poly}} + n_{\text{DMSO}}}, \quad \phi_{\text{Poly}} = \frac{Nn_{\text{Poly}}}{n_{\text{SAM}} + Nn_{\text{Poly}} + n_{\text{DMSO}}}, \quad (4)$$

$$\phi_{\text{DMSO}} = \frac{n_{\text{DMSO}}}{n_{\text{SAM}} + Nn_{\text{Poly}} + n_{\text{DMSO}}}$$

where N is the number of monomers in Poly-4PACz and in our calculations $N = 6$.

MD simulations were performed using GROMACS (version 2019.6)⁴⁴. Three simulation systems were constructed. System 1 consisted of 15 Poly-4PACz molecules, 80 Ph-4PACz molecules and 1,200 DMSO molecules in a cubic box with a side length of ~6 nm. System 2 was built by dispersing 80 Ph-4PACz molecules ~2 nm above the surface of an ITO slab. System 3 was constructed by extracting the Ph-4PACz–Poly-4PACz composite configuration from System 1 and placing it ~2 nm above the ITO surface. For systems 2 and 3, rectangular boxes with dimensions of $7.5 \times 7.2 \times 25 \text{ nm}^3$ were employed to ensure sufficient vacuum space along the z direction. Force field parameters were assigned as follows: ITO was described by the Universal Force Field⁴⁵ and all other molecules were modelled using GAFF2⁴⁶. The partial atomic charges of Ph-4PACz, Poly-4PACz and DMSO were derived using the RESP method⁴⁷. The electrostatic potential used for RESP fitting was obtained at the B3LYP-D3(BJ)/6-311G** level of theory using Gaussian16 (ref. 48), and subsequently fitted using the Multiwfn program (version 3.8(dev))⁴⁹. System 1 was simulated in the NPT (constant number of particles, pressure and temperature) ensemble at 298.15 K and 1 bar for 70 ns. Systems 2 and 3 were simulated in the NVT (constant number of particles, volume and temperature) ensemble at 298.15 K for 5 ns, as structural relaxation occurred rapidly in these cases. The integration time step was set to 1 fs. Electrostatic interactions were computed using the particle mesh Ewald method, while van der Waals interactions were described by the Lennard–Jones potential with a cut-off of 1.2 nm. Cross-interactions were treated using the Lorentz–Berthelot combining rules. Periodic boundary conditions were applied in all dimensions. Molecular structures were visualized using the VMD (Visual Molecular Dynamics) software (version 1.9.3)⁵⁰.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available within the Article and its Supplementary Information. Source data are provided with this paper.

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Author contributions

Y.H., X.W., R.L. and N.L. conceived the idea and designed the experiments. Y.H. supervised the project. X.W. and J.L. fabricated the solar cells and modules. R.L. performed the DFT calculations and theoretical analysis. S.U. and A.N. conducted the lock-in thermography and analysed the data. X.Y. synthesized PolyCz-C4H9. X.W., T.W., Z.Z., Y.W., J.H., J.C., Z.J., Z.S., X.D., X.N., J.A.S., E.S. and Z.D. assisted with the device and materials characterizations. X.W., R.L., N.L., T.W. and Y.H. analysed the data and wrote the paper. Y.H., X.W., R.L., N.L., T.W. and J.L. reviewed and edited the paper. All authors read and commented on the paper.

Competing interests

Y.H. is the founder of Singfilm Solar, a company commercializing perovskite photovoltaics. The remaining authors declare no competing interests.

Additional information

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Solar Cells Reporting Summary

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► Experimental design

Please check: are the following details reported in the manuscript?

1. Dimensions

- Area of the tested solar cells ☒ Yes ☐ No The area of tested solar cell is 0.05 cm² and solar module is 20.25 cm².
- Method used to determine the device area ☒ Yes ☐ No Metal aperture mask (black color)

2. Current-voltage characterization

- Current density-voltage (J-V) plots in both forward and backward direction ☒ Yes ☐ No We provide J-V in both forward and backward direction in Fig. 4e
- Voltage scan conditions ☒ Yes ☐ No See details in Method.
For instance: scan direction, speed, dwell times
- Test environment ☒ Yes ☐ No See details in Method.
For instance: characterization temperature, in air or in glove box
- Protocol for preconditioning of the device before its characterization ☐ Yes ☒ No No preconditioning condition is used.
- Stability of the J-V characteristic ☒ Yes ☐ No We provided MPPT data in Fig.4e
Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see ref. 7 for details.

3. Hysteresis or any other unusual behaviour

- Description of the unusual behaviour observed during the characterization ☐ Yes ☒ No Negligible hysteresis was observed.
- Related experimental data ☒ Yes ☐ No Reverse and forward JV scan can be found in Fig. 4e and Supplementary Fig. 52.

4. Efficiency

- External quantum efficiency (EQE) or incident photons to current efficiency (IPCE) ☒ Yes ☐ No We provide EQE measurement in Supplementary Fig.46
- A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator ☒ Yes ☐ No We compare the integrated Jsc with one from JV scan. The difference between the integrated Jsc from EQE and Jsc from JV scan is within 1.7 % difference, which is within accuracy confidence of the measurements. Integrated Jsc from EQE is higher than Jsc from J-V scan.
- For tandem solar cells, the bias illumination and bias voltage used for each subcell ☐ Yes ☒ No We didn't fabricate tandem solar cell.

5. Calibration

- Light source and reference cell or sensor used for the characterization ☒ Yes ☐ No See details in Method.
- Confirmation that the reference cell was calibrated and certified ☒ Yes ☐ No Our solar simulator was calibrated by Si reference cell.(certificated by Newport.)

Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	The champion devices were sent to do certification, the certified device spectra mismatch factor (0.9954) can be found in Supplementary Fig. 52.
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	Optical aperture masks(0.050 and 20.25 cm ²) were used.
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	No significant variations was observed.
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	We provide certification data in Supplementary Fig. 52.
A copy of any certificate(s) <i>Provide in Supplementary Information</i>	☒ Yes ☐ No	We provide certification data in Supplementary Fig. 52.
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	See details in Fig. 4b, Supplementary Fig. 51
Statistical analysis of the device performance	☒ Yes ☐ No	We provide statistics in Fig.4b, 4c and Supplementary Fig. 43 and 51
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions <i>For instance: illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature</i>	☒ Yes ☐ No	We provided stability data and detailed testing condition Fig.3d, 3e, 3h and.4g